

ENTITY RELATIONSHIP DIAGRAM (ERD)

Working Capital Portfolio Monitoring Dashboard

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1. Introduction

This document defines the Entity Relationship Diagram (ERD) structure for the Working Capital Portfolio Monitoring Dashboard, primarily focused on OD/CC accounts.

The ERD was designed to support:

- Portfolio-level KPI calculations (Page 1)
- Renewal pipeline and interest servicing tracking for Relationship Managers (Page 2)
- Utilization and compliance monitoring (Page 3)
- Product, industry, segment, and RM-wise concentration analysis (Page 4)
- Consolidated, exportable account-level extract for all users (Page 5)
- SQL Server query development
- Power BI dashboard integration

The model follows a star schema — a central fact table (Loan Account) linked to five master/dimension tables and seven transactional/detail tables.

2. ERD Objective

The purpose of the ERD is to:

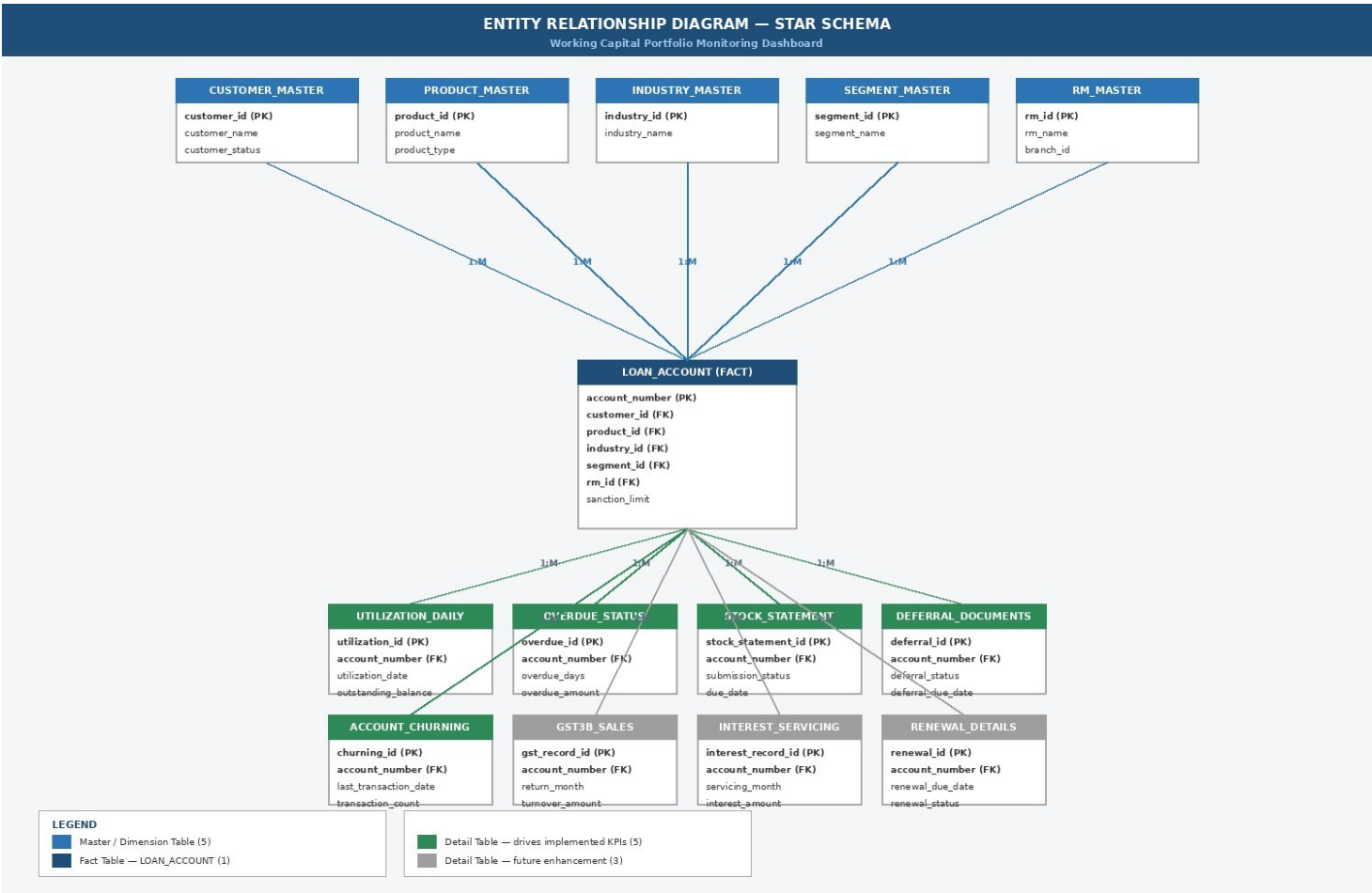
- Organize portfolio data into a logical, query-friendly structure
- Define clear primary key / foreign key relationships
- Support all 19 KPI calculations across the five dashboard pages
- Enable consistent filtering by product, industry, segment, and RM in Power BI
- Keep the model realistic and explainable in interviews

3. Core Entities

Entity	Type	Purpose
CUSTOMER_MASTER	Dimension	Customer identification details
PRODUCT_MASTER	Dimension	Working Capital product types (OD, CC, etc.)
INDUSTRY_MASTER	Dimension	Industry classification for concentration analysis
SEGMENT_MASTER	Dimension	Customer segment classification
RM_MASTER	Dimension	Relationship Manager details
LOAN_ACCOUNT	Fact	Central table — sanctioned limits, account-level keys
OVERDUE_STATUS	Detail	Overdue days — drives SMA/NPA classification
STOCK_STATEMENT	Detail	Stock statement submission tracking
DEFERRAL_DOCUMENTS	Detail	Pending deferral document tracking
ACCOUNT_CHURNING	Detail	Transaction activity — non-churning identification
GST3B_SALES	Detail	GST sales turnover data
INTEREST_SERVICING	Detail	Interest debit / servicing records
RENEWAL_DETAILS	Detail	Account renewal due dates and status

4. ERD Relationship Flow (Star Schema)

LOAN_ACCOUNT is the central fact table (1 record per account). All five master tables relate to LOAN_ACCOUNT in a 1:M relationship (one customer/product/industry/segment/RM → many accounts). All seven detail tables relate to LOAN_ACCOUNT in a 1:M relationship (one account → many daily/periodic records).



5. Entity Field Structures — Dimension Tables

CUSTOMER_MASTER

customer_id (PK)
customer_name
customer_status

PRODUCT_MASTER

product_id (PK)
product_name
product_type

INDUSTRY_MASTER

industry_id (PK)
industry_name

SEGMENT_MASTER

segment_id (PK)
segment_name

RM_MASTER

rm_id (PK)
rm_name
branch_id

6. Entity Field Structure — Fact Table

LOAN_ACCOUNT (FACT)

account_number (PK)
customer_id (FK → CUSTOMER_MASTER)
product_id (FK → PRODUCT_MASTER)
industry_id (FK → INDUSTRY_MASTER)
segment_id (FK → SEGMENT_MASTER)
rm_id (FK → RM_MASTER)
sanction_limit
account_status

7. Entity Field Structures — Detail Tables

All seven detail tables below drive the 19 implemented dashboard KPIs across Portfolio Overview, RM Workbench, Risk & Compliance, and Portfolio Analytics.

OVERDUE_STATUS

overdue_id (PK)
account_number (FK → LOAN_ACCOUNT)
overdue_days
overdue_amount

STOCK_STATEMENT

stock_statement_id (PK)
account_number (FK → LOAN_ACCOUNT)
statement_month
submission_status
due_date

DEFERRAL_DOCUMENTS

deferral_id (PK)
account_number (FK → LOAN_ACCOUNT)
document_type
deferral_status
deferral_due_date

ACCOUNT_CHURNING

churning_id (PK)
account_number (FK → LOAN_ACCOUNT)
last_transaction_date
transaction_count

GST3B_SALES

gst_record_id (PK)
account_number (FK → LOAN_ACCOUNT)
return_month
turnover_amount

INTEREST_SERVICING

interest_record_id (PK)
account_number (FK → LOAN_ACCOUNT)
servicing_month
interest_amount
servicing_status

RENEWAL_DETAILS

renewal_id (PK)

account_number (FK → LOAN_ACCOUNT)
last_renewal_date
renewal_due_date
renewal_status

8. Business Logic Supported by ERD

Business Capability	Supporting Entities	KPI Reference
Total Portfolio Visibility	CUSTOMER_MASTER, LOAN_ACCOUNT	KPI-01, KPI-02
Utilization Monitoring	LOAN_ACCOUNT	KPI-03, KPI-04, KPI-07, KPI-08
SMA / NPA Classification	LOAN_ACCOUNT, OVERDUE_STATUS	KPI-05, KPI-06
Stock Statement Compliance	LOAN_ACCOUNT, STOCK_STATEMENT	KPI-09
Deferral Tracking	LOAN_ACCOUNT, DEFERRAL_DOCUMENTS	KPI-10
Account Churning Identification	LOAN_ACCOUNT, ACCOUNT_CHURNING	KPI-11
Product-wise Analysis	LOAN_ACCOUNT, PRODUCT_MASTER	KPI-12
Industry-wise Concentration	LOAN_ACCOUNT, INDUSTRY_MASTER	KPI-13
Segment-wise Analysis	LOAN_ACCOUNT, SEGMENT_MASTER	KPI-14
RM-wise Distribution	LOAN_ACCOUNT, RM_MASTER	KPI-15
Renewal Monitoring	LOAN_ACCOUNT, RENEWAL_DETAILS	KPI-16, KPI-17 (Page 2)
Cash Flow / Turnover Monitoring	LOAN_ACCOUNT, GST3B_SALES	Risk & Compliance line chart — early-warning indicator
Interest Servicing Monitoring	LOAN_ACCOUNT, INTEREST_SERVICING	KPI-18 (Page 2)

9. Key Design Highlights

The ERD was designed to support:

- A star schema with LOAN_ACCOUNT as the central fact table
- Five master / dimension tables enabling consistent filtering across all dashboard pages
- Seven detail tables supporting compliance, utilization, and operational KPIs
- All 19 implemented KPIs traceable to specific entities and fields
- Three of the seven detail tables (RENEWAL_DETAILS, GST3B_SALES, INTEREST_SERVICING) support RM Workbench and Risk & Compliance monitoring — renewal pipeline, interest servicing status, and cash-flow early-warning trend — rather than being future-only additions

The structure keeps the data model practical, normalized, and easy to explain in interviews — a single fact table with clean star-schema relationships to master and detail tables.

10. Conclusion

The ERD provides a star-schema analytical data model for the Working Capital Portfolio Monitoring Dashboard, comprising 13 tables: 5 master/dimension tables, 1 central fact table (Loan Account), and 7 detail tables.

The structure supports:

- All 19 KPIs across Portfolio Overview, RM Workbench, Risk & Compliance, Portfolio Analytics, and Data Export
- Clean, traceable SQL joins from LOAN_ACCOUNT to every master and detail table
- Consistent filtering and drill-down in Power BI by product, industry, segment, and RM
- A realistic, interview-ready data model that mirrors how working capital portfolios are structured in core banking systems